Arista Access Node Diagnostic and Resolution SOP for Minor Optical Transport Failure (HEB-ACC-43)

Node: HEB-ACC-43 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9664

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a Minor Optical Transport Failure (Alarm Code: ALARM_9664) on the Arista Access Node (Node ID: HEB-ACC-43) located at the Hebbal Edge site. The failure domain in question is attributed to the degradation of the SFP/XFP, fiber cuts, or Rx/Tx power loss within the Optical Transport Network (OTN). The Arista Access Node, being a crucial component of the OTN, is responsible for transmitting and receiving optical signals. In this context, the failure of the SFP/XFP or fiber cuts can lead to a loss of signal quality, resulting in the aforementioned alarm code. The OTN operates based on the Optical Transport Network (OTN) protocol, which utilizes the Transport overhead bytes to transmit control and management information. The failure of the OTN protocol can be attributed to the degradation of the SFP/XFP or fiber cuts, leading to a loss of synchronization between the transmitter and receiver, resulting in the activation of the alarm. This SOP will guide the engineer through the diagnostic and resolution steps to rectify the issue and restore the normal functioning of the Arista Access Node.

Pre-requisite Checks

- check: Verify the site location and node ID to ensure that the issue is specific to HEB-ACC-43.

- check: Confirm the severity of the alarm and ensure that it is Minor (ALARM_9664).

- check: Conduct a physical inspection of the SFP/XFP and fiber connections to identify any signs of degradation or damage.

- check: Verify the status of the redundant link to ensure that it is functioning properly.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface e1-1/1/1, Expected Output: Interface e1-1/1/1 is up, line protocol is up, Description: This command verifies the status of the interface and ensures that the line protocol is up, indicating that the interface is functioning properly.

Command: show sfp e1-1/1/1, Expected Output: SFP e1-1/1/1 is present and operational, Description: This command verifies the presence and
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operational status of the SFP/XFP, indicating whether it is
functioning properly or not.

Command: show optical tx e1-1/1/1, Expected Output: Optical TX power
is within the normal range, Description: This command verifies the
optical TX power to ensure that it is within the normal range,
indicating that the transmitter is functioning properly.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: no sfp e1-1/1/1, Description: This command removes the existing SFP/XFP and prepares the interface for a new one.

Step 2: Step: Verify the new SFP/XFP, Command: show sfp e1-1/1/1, Description: This command verifies the presence and operational status of the new SFP/XFP, indicating whether it is functioning properly or not.

Step 3: Step: Verify the optical TX power, Command: show optical tx e1-1/1/1, Description: This command verifies the optical TX power to ensure that it is within the normal range, indicating that the transmitter is functioning properly.

Post-Check Verification

```
Command: show interface e1-1/1/1, Expected Output: Interface e1-1/1/1
is up, line protocol is up, Description: This command verifies the
status of the interface and ensures that the line protocol is up,
indicating that the interface is functioning properly.

Command: show sfp e1-1/1/1, Expected Output: SFP e1-1/1/1 is present
and operational, Description: This command verifies the presence and
operational status of the SFP/XFP, indicating whether it is
functioning properly or not.

Command: show optical tx e1-1/1/1, Expected Output: Optical TX power
is within the normal range, Description: This command verifies the
optical TX power to ensure that it is within the normal range,
indicating that the transmitter is functioning properly.
```

Roll-back Procedures

- Step: Reinstall the original SFP/XFP, Command: sfp e1-1/1/1 original-sfp, Description: This command reinstalls the original SFP/XFP, restoring the interface to its previous state.

- Step: Verify the interface status, Command: show interface e1-1/1/1, Description: This command verifies the status of the interface and ensures that the line protocol is up, indicating that the interface is functioning properly.